

RESUME

Hash Nabi

Researcher & Technical Program Manager — Economics, Mathematics, Protocol Design, Governance, Systems Engineering
coad1024@gmail.com

Value Proposition

I design, analyze, and lead the development of **complex socio-technical economic systems**. With a strong background in Aerospace Engineering my work integrates **mathematics, economic/game-theoretic modeling, governance architecture, mechanism design, and systems engineering** with **cross-functional leadership** and **strategic communication**. I thrive in **high-autonomy, async-first environments**, bridging research, engineering, product, and governance to deliver **decision-ready insights** for top Web3 protocols.

Professional Experience

Independent Research Contractor — BlockScience (2018 – Present)

Operate at the intersection of systems engineering, economic mechanism design, and governance analysis, flexing between researcher and technical program manager roles.

Key Projects:

- **Avalanche Economics Specifications** (*Research Lead*) — MBSE mapping of entire Avalanche ecosystem role taxonomy, mechanisms, consensus, L1 economics; cadCAD simulations for validator incentives and treasury mechanisms.
- **OlympusDAO — Range Bound Stability Simulation** (*TPM & Researcher*) — Led multi-disciplinary team; modeled stability mechanism under stress scenarios; delivered extensible simulation infrastructure.
- **Hedera Governance Benchmarking** (*TPM & Researcher*) — Coordinated governance benchmarking study; analyzed council-based governance vs. DAO models; surfaced decentralization trade-offs.

- **Aztec Sequencer Selection** (*TPM & Researcher*) — Scoped and managed comparative analysis of sequencer mechanisms; integrated technical, economic, and governance perspectives.
 - **Protocol Labs Governance Study & Filecoin Consensus Pledge** (*Technical Writer & Research Lead*) — Directed governance benchmarking research; co-authored *Understanding the Filecoin Consensus Pledge*, explaining mechanism design, security, and liquidity implications.
 - **Holo Reserve Account Mechanism** (*TPM & Contributor*) — Coordinated engineers, economists, and data scientists; contributed to regulatory brief on price discovery algorithms and reserve account design.
 - **Bonding Curve Research Group Library** (*Founder, Writer & Researcher*) — Built open-access library synthesizing bonding curve theory, design patterns, and economic implications; established shared conceptual vocabulary for the primitive design.
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Education

- **MSc Economics** — Indian Institute of Technology, Delhi
 - **BEng Aerospace Engineering** — School of Aerospace Engineering, Universiti Sains Malaysia
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Selected Publications & Outputs

- [Understanding Large-Language Models, LLM System Foundations](#) — BlockScience Blog
- [Simulating the Heights of Olympus](#) — BlockScience Blog
- [Bonding Curve Research Group Library](#) — GitBook
- [Understanding the Filecoin Consensus Pledge](#) — BlockScience Blog
- [Exploring the Filecoin Termination Fee](#) — BlockScience Blog
- [Intro to Avalanche Economic Specification](#) — HackMD
- [Avalanche Subsystem Specifications](#) — HackMD
- [Avalanche Differential Specifications](#) — HackMD
- [Engineering Complex Adaptive Systems](#) — Medium
- [Mapping the Computer-Aided Governance Process](#) — Medium
- [Introducing Automated Regression Market Makers \(ARMMs\): A New Price Discovery Mechanism for Semi-Fungible Assets](#) — BlockScience Blog
- [From Nash to Lyapunov](#) — BlockScience Blog
- [Complex Systems Engineering | Digital Infrastructure](#) — Medium
- [Conceptual Models for DAO2DAO Relations](#) — Medium
- [Introducing Govbase](#) — Medium

Skills Heat Map

Category	Skill	Grade	Heat
Core Technical	Smart Contract Literacy & Rapid Prototyping	B-	
	DeFi Mechanism & Layer-2 Ecosystems Knowledge	A	
	Model-Based Systems Engineering (MBSE) & Economic Security Awareness	B+	
	Privacy Tech	C+	
	Technical Program Management (TPM)	A+	
	Knowledge Infrastructure & AI-Powered Pipelines	A	
Research & Analysis	Dissect Whitepapers/Specs	A	
	Benchmark Protocols	A	
	Tokenomics/Game Theory	B+	
Communication	Technical Writing & Context Framing	A	
	Cross-Team Translation	A+	

Cultural Fit	Self-Directed Exploration & Async Collaboration	A+	
	Ecosystem Immersion	B+	

Cover Letter

Dear Hiring Manager,

I am excited to apply for the **Researcher position** at Wonderland. My career has been defined by designing, analyzing, and leading the development of complex socio-technical economic systems for some of the most innovative protocols in Web3.

At BlockScience, I have operated in a multi-role capacity — researcher, social system engineer, governance analyst, engineering communicator & technical writer and technical program manager — delivering high-impact work such as:

- **Avalanche Economics Specifications** — Applied MBSE to map consensus, staking, and subnet economics; built a computable economic twin to test validator incentives and treasury mechanisms.
- **OlympusDAO RBS Simulation** (*TPM & Researcher*) — Led a multi-disciplinary team to model and stress-test stability mechanisms, producing extensible simulation infrastructure.
- **Hedera Governance Benchmarking** (*TPM & Researcher*) — Coordinated a comparative governance study, surfacing trade-offs in decentralization and decision-making.
- **Aztec Sequencer Selection** (*TPM & Researcher*) — Integrated technical, economic, and governance perspectives into a unified evaluation framework.
- **Protocol Labs & Filecoin** (*Technical Writer & Research Lead*) — Directed governance benchmarking research and co-authored *Understanding the Filecoin Consensus Pledge*, translating complex economic mechanisms into decision-ready insights.
- **Holo Reserve Account Mechanism** (*TPM & Contributor*) — Coordinated engineers, economists, and data scientists to produce a regulatory brief on price discovery algorithms and reserve account design.

- **Bonding Curve Research Group** (*Writer & Researcher*) — Founded and maintain an open-access library that codifies bonding curve theory and design patterns for the broader ecosystem.

Across these projects, I have demonstrated the ability to **adapt dynamically to scope**, lead **multi-disciplinary teams**, and **translate complex modeling into actionable strategy**. My dual background in aerospace engineering and economics gives me both the analytical rigor and the systems thinking needed to navigate Wonderland's interdisciplinary challenges.

I am drawn to Wonderland's mission because it aligns with my own — to design mechanisms and governance structures that are both economically sound and socially sustainable. I would welcome the opportunity to bring my experience in **mechanism design, governance benchmarking, and technical program leadership** to your research team.

Sincerely,
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